National University of Computer & Emerging Sciences- Karachi campus

Applied Human Computer Interaction (CS3014) **Sessional II Exam**

Course instructors: **Total time: 1 hour**

Mr. Shaheer, Mr. Kashan Hussain, Mr. Ghulam Murtaza, **Total marks:40**

Mr. Asif Fazwani and Mr. Jibran Rasheed Khan **Total Questions:4**

NOTE: Attempt all questions in order, out of order answers will not be marked

CLO 2: Apply user-centered design principles to design interactive interfaces

Q1) A agricultural tech company installs **FarmSense**, a tablet-based system for farmers to monitor soil moisture, weather forecasts, crop spraying schedules, and equipment maintenance. The system is used outdoors in bright sunlight.

Issues observed:

- The screen glare makes text unreadable in direct sunlight.
- Farmers must navigate through 5 menus to log a completed spraying task.
- The system uses red text for both critical alerts (e.g., pest outbreak) and routine reminders (e.g., refill fuel).
- No undo button exists for accidental task deletion.
- Elderly farmers forget the multi-step process to view weather history.
- The tablet requires precise finger taps, difficult with work gloves.

Answer the following:
a. Identify ergonomic and visual design issues in FarmSense.
b. What is "design inertia"? How could it harm FarmSense?

CLO3: Analyze usability issues and evaluate interfaces using standard HCI techniques
Q2) Discuss the ways in which the full- page word processor is or is not a direct manipulation interface for editing a document using Schneiderman's criteria. What features of a modern word processor break the metaphor of composition with pen (or typewriter), and paper?

CLO3: Analyze usability issues and evaluate interfaces using standard HCI techniques
Q3) A software company is developing a **hospital management system**. The system is intended to help doctors and staff manage patient records, prescriptions, and emergency alerts.

During development:

- The team strictly follows the **initial requirements document** and ensures that every function matches the specification exactly.

- However, after deployment, doctors complain that the system is **difficult to use** and does not fit their real workflow.
- The designers realize that although the system was technically correct, it did not meet user expectations.
- There was also a noticeable **gap between formal design models (technical diagrams, specifications)** and how users actually interact with the system.

Based on the above scenario:

a. Identify whether the issue is related to **verification or validation**, and justify your answer.
b. Explain the concept of the **formality gap** and how it appears in this scenario.
c. Recommend whether a **linear model** or an **iterative model** should be used, with justification.

CLO3: Analyze usability issues and evaluate interfaces using standard HCI techniques
Q4) A product team is building a feature for a mobile wallet app called **FASTPAY**, allowing users to send money instantly.

Usability Goals

- Send money in ≤ 5 seconds
- Max 2 taps to complete transaction
- "Cancel transaction" available in 1 tap (within 3 seconds)
- 98% task completion rate

Test Results

- 99% users successfully sent money
- Avg time = 4 seconds
- Cancel option works in 1 tap

User Feedback

- "I sent money to the wrong person by my mistake"
- "Due to lack of knowledge, I didn't get time to confirm details"

a) Which ISO 9241 usability standard dimensions are satisfied? Which dimension is clearly failing and Why?

b) What is the main usability issue in this design? Is speed always a good usability goal? Explain.

c). What trade-off exists between efficiency and error prevention in this scenario? How did the design prioritize one at the cost of another?